

Problem 1:

1-a)

$$\min_{x_1, x_2} x_1 + x_2, \quad \text{s.t. } x_1 \geq -2, \quad x_1 + 2x_2 \geq -2$$

The constraints can be rewritten as:

$$g_1(x) = x_1 + 2 \geq 0 \rightarrow -x_1 + 0x_2 - 2 \leq 0$$

$$g_2(x) = x_1 + 2x_2 + 2 \geq 0 \rightarrow -x_1 - 2x_2 - 2 \leq 0$$

In matrix form:

$$[-1, 0; -1, -2] * [x_1; x_2] + [-2; -2] \leq 0$$

1-b)

$$L(x, \lambda) = (x_1 + x_2) + \lambda_1(-x_1 - 2) + \lambda_2(-x_1 - 2x_2 - 2)$$

1-c)

Stationarity:

$$\partial L / \partial x_1 = 0 = 1 - \lambda_1 - \lambda_2$$

$$\partial L / \partial x_2 = 0 = 1 - 2\lambda_2$$

Primal feasibility:

$$g(x^*) \leq 0$$

Leading to

$$-x_1 + 0x_2 - 2 \leq 0$$

$$-x_1 - 2x_2 - 2 \leq 0$$

Dual feasibility:

$$\lambda_1, \lambda_2 \geq 0$$

Complementary slackness:

$$\lambda_i g_i(x^*) = 0$$

Leading to:

$$\lambda_1(-x_1 - 2) = 0$$

$$\lambda_2(-x_1 - 2x_2 - 2) = 0$$

1-d)

Using stationarity:

$$1 - 2\lambda_2 = 0 \rightarrow \lambda_2 = 1/2$$

$$1 - \lambda_1 - \lambda_2 = 0 \rightarrow \lambda_1 = 1/2$$

Plugging into complementary slackness:

$$\lambda_1(-x_1 - 2) = 0 \rightarrow x_1 = -2$$

$$\lambda_2(-x_1 - 2x_2 - 2) = 0 \rightarrow x_2 = 0$$

Checking with primal feasibility:

$$-x_1 + 0x_2 - 2 \leq 0 \rightarrow 2 - 2 \leq 0 \rightarrow \text{satisfied}$$

$$-x_1 - 2x_2 - 2 \leq 0 \rightarrow 2 - 0 - 2 \leq 0 \rightarrow \text{satisfied}$$

Dual feasibility is satisfied since both lambdas are greater than 0

Thus:

$$x_1 = -2$$

$$x_2 = 0$$

1-e)

Both constraints are active and satisfied, since the primal feasibility constraints satisfy an equality, rather than just an inequality

Problem 2)

2-a)

$$\min_{x_1, x_2} (x_1^2 + x_2^2)/2 - 2x_1 - x_2, \text{ s.t. } x_1 + x_2 \leq 1, x_1 \geq 0, x_2 \geq 0$$

Using the partial derivatives:

$$\partial f / \partial x_1 (x_1^2 + x_2^2)/2 - 2x_1 - x_2 = 0 \rightarrow x_1 - 2 = 0 \rightarrow x_1 = 2$$

$$\partial f / \partial x_2 (x_1^2 + x_2^2)/2 - 2x_1 - x_2 = 0 \rightarrow x_2 - 1 = 0 \rightarrow x_2 = 1$$

This satisfies $x_1 \geq 0, x_2 \geq 0$, but not $x_1 + x_2 \leq 1$, so it is not feasible

2-b)

$$g_1(x) = x_1 + x_2 - 1 \leq 0$$

$$g_2(x) = -x_1 \leq 0$$

$$g_3(x) = -x_2 \leq 0$$

2-c)

$$L(x, \lambda) = ((x_1^2 + x_2^2)/2 - 2x_1 - x_2) + \lambda_1(x_1 + x_2 - 1) + \lambda_2(-x_1) + \lambda_3(-x_2)$$

2-d)

Stationarity:

$$\partial L / \partial x_1 = 0 = x_1 - 2 + \lambda_1 - \lambda_2$$

$$\partial L / \partial x_2 = 0 = x_2 - 1 + \lambda_1 - \lambda_3$$

Primal feasibility:

$$g(x^*) \leq 0$$

Leading to

$$x_1 + x_2 - 1 \leq 0$$

$$-x_1 \leq 0$$

$$-x_2 \leq 0$$

Dual feasibility:

$$\lambda_1, \lambda_2, \lambda_3 \geq 0$$

Complementary slackness:

$$\lambda_i g_i(x^*) = 0$$

Leading to:

$$\lambda_1(x_1 + x_2 - 1) = 0$$

$$\lambda_2(-x_1) = 0$$

$$\lambda_3(-x_2) = 0$$

2-e)

Only the condition for g_1 is not satisfied. Set $\lambda_2, \lambda_3 = 0$, and use the stationarity condition for λ_1

$$\partial L / \partial x_1 = 0 = x_1 - 2 + \lambda_1 - \lambda_2 \rightarrow x_1 = 2 - \lambda_1$$

$$\partial L / \partial x_2 = 0 = x_2 - 1 + \lambda_1 - \lambda_3 \rightarrow x_2 = 1 - \lambda_1$$

Using the primal feasibility constraint with an equality, since this is the constraint which is not currently being satisfied:

$$x_1 + x_2 - 1 \leq 0 \rightarrow (2 - \lambda_1) + (1 - \lambda_1) - 1 \leq 0 \rightarrow \lambda_1 = 1$$

This leads to:

$$x_1 = 1, x_2 = 0$$

This satisfies all constraints

2-f)

The unconstrained minimizer is different from the constrained one, because the unconstrained one violates some of the constraints. In order to ensure that all constraints are satisfied, the unconstrained solution must be pulled until it just barely satisfies the g_1 constraint. This is the result of using the stationarity condition with primal feasibility.

Problem 3)

3-a,b)

The area is the length of each side multiplied. The combined perimeter must be less than a variable L and all lengths must be non-negative. This can be written as:

$$\min_{x_1, x_2} -x_1 * x_2, \text{ s.t. } 2x_1 + 2x_2 \leq l, x_1 \geq 0, x_2 \geq 0$$

3-c)

$$g_1(x) = 2x_1 + 2x_2 - l \leq 0$$

$$g_2(x) = -x_1 \leq 0$$

$$g_3(x) = -x_2 \leq 0$$

$$L(x, \lambda) = (-x_1 * x_2) + \lambda_1(2x_1 + 2x_2 - l) + \lambda_2(-x_1) + \lambda_3(-x_2)$$

3-d)

Stationarity:

$$\partial L / \partial x_1 = 0 = -x_2 + 2\lambda_1 - \lambda_2$$

$$\partial L / \partial x_2 = 0 = -x_1 + 2\lambda_1 - \lambda_3$$

Primal feasibility:

$$g(x^*) \leq 0$$

Leading to

$$2x_1 + 2x_2 - l \leq 0$$

$$-x_1 \leq 0$$

$$-x_2 \leq 0$$

Dual feasibility:

$$\lambda_1, \lambda_2, \lambda_3 \geq 0$$

Complementary slackness:

$$\lambda_i g_i(x^*) = 0$$

Leading to:

$$\lambda_1(2x_1 + 2x_2 - l) = 0$$

$$\lambda_2(-x_1) = 0$$

$$\lambda_3(-x_2) = 0$$

3-e)

Satisfying the positivity constraints on λ_2, λ_3 is simple, so let those be 0, unless otherwise shown to be violated. The active condition is condition g_1 on λ_1 , so that will be set to an equality constraint. Using the stationarity and primal feasibility conditions:

$$\partial L / \partial x_1 = 0 = -x_2 + 2\lambda_1 - \lambda_2 \rightarrow x_2 = 2\lambda_1$$

$$\partial L / \partial x_2 = 0 = -x_1 + 2\lambda_1 - \lambda_3 \rightarrow x_1 = 2\lambda_1$$

$$2x_1 + 2x_2 - l \leq 0 \rightarrow 2(2\lambda_1) + 2(2\lambda_1) - l \leq 0 \rightarrow 8\lambda_1 = l \rightarrow \lambda_1 = l/8$$

This yields:

$$x_1 = x_2 = 2\lambda_1 = l/4$$

For positive values of L, all conditions are satisfied.

3-f)

The shape of the optimal rectangle is a square with an area of:

$$A = x_1 x_2 = (l/4)(l/4) = l^2/16$$